SPECIFICATION

Background of the Invention

Cloud computing and virtualization provide resource flexibility by creating virtual machines, which are software-based slices of a single, monolithic physical server. While this allows for rapid provisioning, it introduces performance overhead from the hypervisor layer. Provisioning a true "bare-metal" server, by contrast, is a slow, manual process involving physical hardware allocation. There is a need for a method that combines the speed and flexibility of cloud provisioning with the performance of bare-metal hardware.

Summary of the Invention

The present invention is a method for the dynamic composition of bare-metal computing resources, which leverages a physically disaggregated hardware architecture and an optical data fabric switch. This method provides "bare-metal-as-a-service," where a user can request a specific hardware configuration and have it provisioned on-demand.

The process is managed by a software control plane. It begins when a user or automated orchestrator sends a request to the control plane for a computer with specific attributes (e.g., 8 CPU cores, 1TB of RAM, 2 GPUs). The control plane consults an inventory of available hardware in the disaggregated resource pools (CPU pool, memory pool, GPU pool) and selects the necessary components. It then sends a series of commands to an optical data fabric switch, instructing it to create the specific point-to-point optical links that would connect these selected components as if they were on a single motherboard. The resulting logical computer is then provisioned to the user. When the user's task is complete, the control plane instructs the switch to dissolve the links, and the individual hardware components are returned to their respective pools, available for the next composition request.

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Brief Description of the Drawings

FIG. 1 is a flowchart illustrating the steps of the dynamic composition and decomposition method.

Detailed Description of the Invention

Referring to FIG. 1, the method is shown as a sequence of steps. The process begins with a control plane receiving a request for a bare-metal computer with specified resources (101). The control plane identifies and reserves the required available components from the disaggregated CPU, memory, and other resource pools (102). It then sends a command to the optical data fabric switch (103), instructing it to establish the necessary optical links to interconnect the reserved components. The switch executes the command, creating a logical computer (104). This newly composed computer is then provisioned to the user for a finite duration (105). Once the task is complete and the user releases the machine, the control plane sends a teardown command to the switch (106), which dissolves the optical links. Finally, the individual hardware components are returned to the available resource pools (107).